Mobile Robotics (ME6106) | Semester Design Project | IIT Patna

Autonomous Warehouse Challenge 2026

Engineering an Autonomous Mobile Robot from Concept to Deployment

1. Challenge Overview

Modern warehouses increasingly rely on **Autonomous Mobile Robots (AMRs)** to transport goods, replenish inventory, inspect facilities, and collaborate safely with human workers. Unlike robots operating in structured industrial environments, warehouse robots must function under uncertainty—dealing with imperfect sensing, changing layouts, moving obstacles, localization errors, and unexpected events.



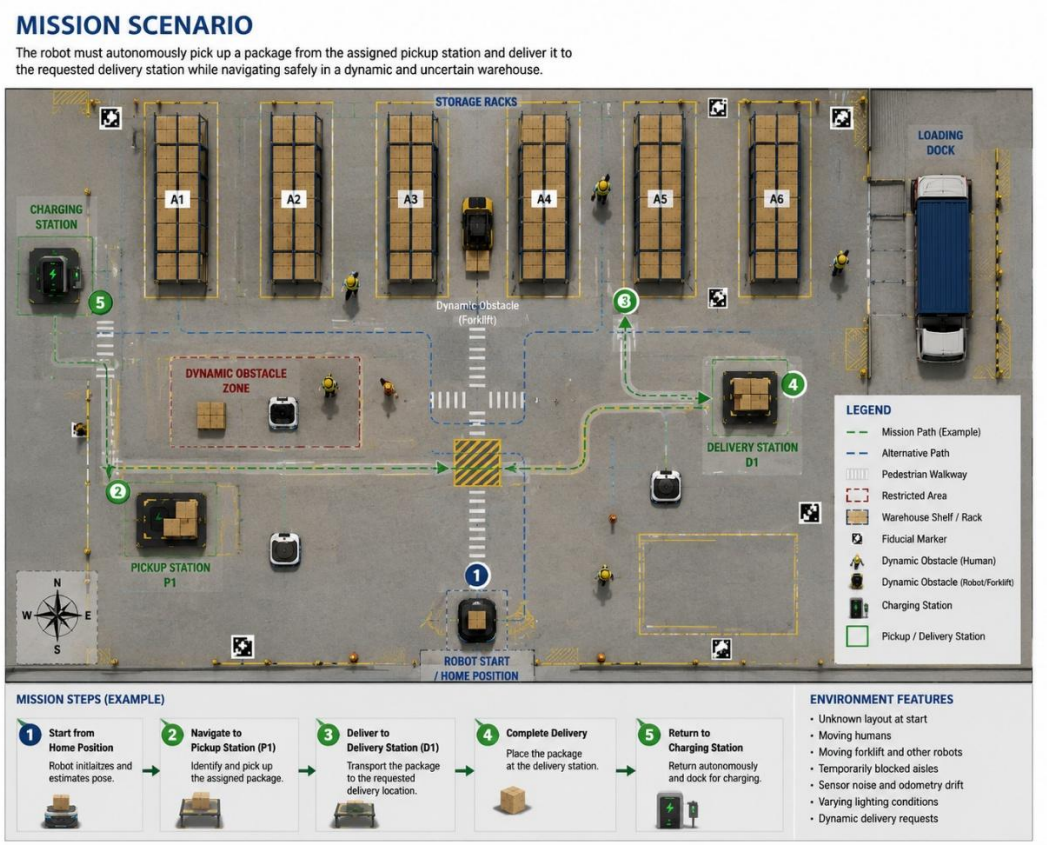
The objective of this semester project is to **design, simulate, and evaluate an autonomous robotic system capable of performing logistics operations in a realistic warehouse environment developed in CoppeliaSim**.

Unlike traditional robotics assignments where every team uses the same robot, **each team is responsible for designing its own robotic platform**, selecting an appropriate locomotion mechanism, justifying its engineering decisions, and developing the complete autonomy stack.

The emphasis of this project is not only on programming autonomous behavior but also on making informed engineering choices regarding robot morphology, sensing, navigation, and system integration.

2. Mission Scenario

A warehouse receives continuous requests to transport packages between storage racks, loading docks, inspection stations, and delivery points.



Your team has been contracted to develop an autonomous warehouse robot capable of

- transporting packages,
- navigating safely,
- operating continuously,
- avoiding obstacles,
- recovering from disturbances,
- adapting to uncertain environments.

Every delivery request is generated dynamically during execution.

The robot must complete its mission without human intervention.

3. Engineering Design Challenge

Unlike commercial robotic platforms, **no robot architecture is prescribed**.

Each team shall answer the following engineering question:

What is the most suitable robot for this warehouse?

This decision should be supported using engineering analysis rather than intuition.

Teams are expected to evaluate alternative locomotion mechanisms and justify why their chosen design is appropriate for the warehouse mission.

4. Robot Design Requirements

Each team shall design and simulate its own robotic platform.

The design process should include:

- mission analysis,
- functional decomposition,
- concept generation,
- trade-off analysis,
- final concept selection,
- kinematic modeling,
- sensor integration,
- autonomy implementation.

The final robot should be optimized for warehouse operation.

Permitted Locomotion Platforms

Teams may choose one of the following or propose an equivalent design.

- Differential-drive robot
- Tricycle robot
- Ackermann-steered robot
- Four-wheel steering robot
- Skid-steer robot
- Omniwheel robot
- Mecanum-wheel robot

- Quadruped robot
- Hexapod robot
- Snake robot
- Hybrid wheel–leg robot
- Bio-inspired custom design (subject to instructor approval)

The chosen locomotion mechanism must be justified through engineering analysis.

Robot Design Justification

Each team shall compare at least three candidate concepts using a structured decision matrix.

Typical evaluation criteria include

- maneuverability,
- mobility,
- stability,
- turning radius,
- payload capacity,
- mechanical complexity,
- controllability,
- energy efficiency,
- robustness,
- manufacturability,
- suitability for warehouse logistics.

Required Robot Subsystems

Each robot shall include

Mobility System

- locomotion mechanism
- wheel/leg configuration

- actuation

Perception System

At least three sensors selected appropriately from

- wheel encoders,
- LiDAR,
- RGB camera,
- depth camera,
- IMU,
- ultrasonic sensors,
- proximity sensors,
- fiducial marker detection.

Control System

- motion controller,
- trajectory controller,
- emergency stop,
- obstacle avoidance.

Autonomy System

- localization,
- mapping,
- path planning,
- task planning.

5. Warehouse Environment

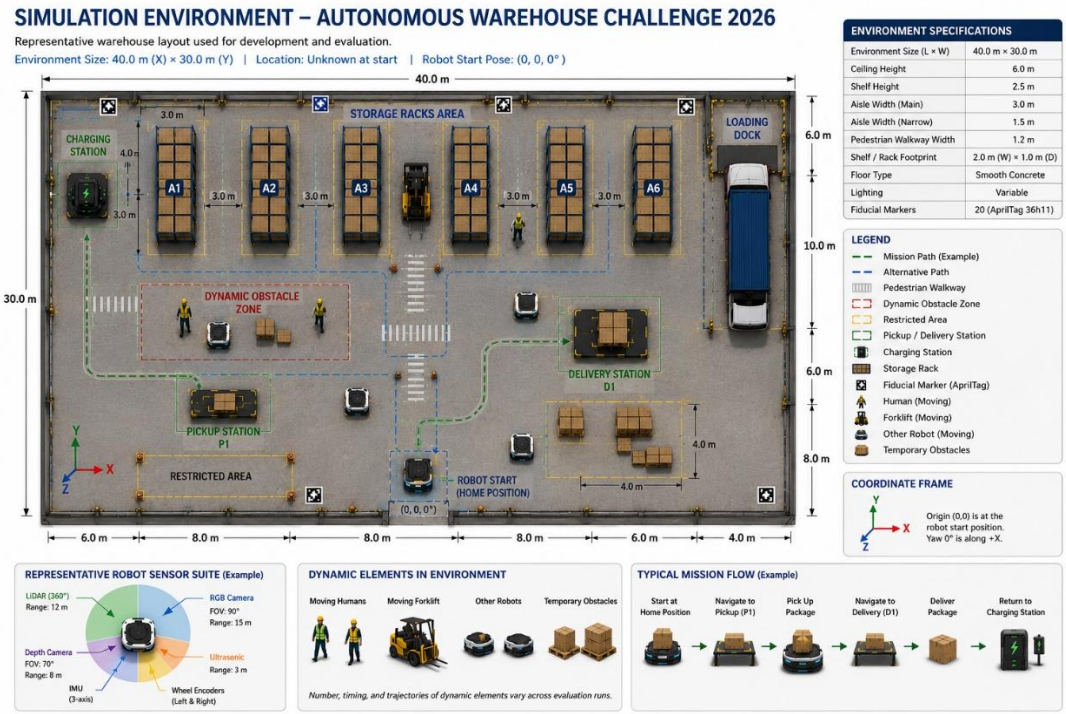
The warehouse contains

- storage racks,
- pickup stations,
- loading docks,
- charging stations,

- delivery stations,
- narrow aisles,
- intersections,
- moving workers,
- forklifts,
- autonomous robots,
- temporary obstacles.

The exact layout is intentionally unknown at the beginning of the project. A representative picture of the environment has been provided below.

Different evaluation scenarios may contain different warehouse layouts.



6. Mission Tasks

The autonomous robot shall

1. initialize,
2. estimate its pose,
3. navigate to pickup,
4. identify the package,

5. transport the package,
6. avoid collisions,
7. replan if necessary,
8. deliver the package,
9. return to the charging station,
10. repeat for multiple delivery requests.

7. Uncertainty Model

The project explicitly evaluates robot performance under uncertainty.

Robot Uncertainty

- wheel slip,
- odometry drift,
- encoder noise,
- IMU bias,
- actuator saturation,
- controller delay,
- camera noise,
- LiDAR measurement noise.

Environmental Uncertainty

- unknown warehouse layout,
- blocked aisles,
- moving humans,
- moving forklifts,
- temporary obstacles,
- changing package locations,
- varying illumination,
- unexpected delivery requests.

Students are expected to discuss how these uncertainties influence the behavior of their system and what mitigation strategies have been implemented.

8. Project Phases

Phase I — Robot Design and Mobility

Objective: Engineer a robot platform suitable for the warehouse mission.

Tasks

- Analyze mission requirements.
- Generate and compare multiple robot concepts.
- Select a locomotion platform using a trade-off analysis.
- Develop the robot model in CoppeliaSim.
- Derive and validate the robot's kinematic model.
- Demonstrate basic teleoperation and waypoint following.

Deliverables

- Design report with concept selection.
- Trade-off matrix.
- CAD/CoppeliaSim model.
- Kinematic analysis.
- Demonstration video.
- Presentation.

Phase II — Perception and Environmental Awareness

Objective: Enable the robot to perceive and understand its surroundings.

Tasks

- Integrate the selected sensor suite.
- Acquire and visualize sensor data.
- Detect obstacles, aisles, shelves, and packages.
- Implement emergency obstacle avoidance.

- Evaluate sensor performance under noise.

Deliverables

- Sensor integration report.
- Perception software.
- Demonstration.
- Presentation.

Phase III — Localization, Mapping, and Navigation

Objective: Achieve reliable autonomous navigation.

Tasks

- Develop occupancy-grid or equivalent maps.
- Implement localization.
- Perform global path planning.
- Perform local obstacle avoidance.
- Implement dynamic replanning.
- Validate navigation under uncertainty.

Deliverables

- Autonomous navigation demonstration.
- Mapping results.
- Localization accuracy analysis.
- Technical report.
- Presentation.

Phase IV — Integrated Warehouse Challenge

Objective: Demonstrate a fully autonomous logistics robot.

Tasks

- Receive delivery requests.
- Navigate autonomously.

- Pick up and deliver packages.
- Avoid moving obstacles.
- Recover from blocked paths.
- Handle multiple delivery missions.
- Return to charging station.
- Collect and report performance metrics.

Deliverables

- Live demonstration.
- Final technical report.
- Code repository.
- User manual.
- Final presentation.

9. Key Performance Metrics

Evaluation will focus on engineering quality, autonomy, and robustness rather than a single "winner." The following metrics will be used:

Category	Weight	Representative Metrics
Robot Design & Engineering Justification	**15%**	Mission analysis, concept generation, trade-off study, suitability of chosen locomotion platform
Locomotion Performance	**15%**	Mobility, maneuverability, stability, turning performance, trajectory tracking accuracy
Perception & Environmental Awareness	**10%**	Sensor integration, detection accuracy, robustness to sensor noise
Localization & Mapping	**10%**	Pose estimation accuracy, map quality, resilience to odometry drift
Navigation & Motion Planning	**20%**	Path efficiency, obstacle avoidance, replanning capability, navigation success rate

Autonomous Mission Execution	**20%**	Successful deliveries, task completion rate, collision-free operation, mission time
Robustness to Uncertainty	**5%**	Recovery from blocked aisles, dynamic obstacles, localization failures, sensor degradation
Software Engineering & Documentation	**5%**	Code organization, modularity, reproducibility, technical documentation

10. Bonus Challenges

Teams seeking additional distinction may implement advanced features such as:

- Multi-robot fleet coordination.
- Autonomous task allocation.
- Human-aware navigation.
- Semantic mapping.
- SLAM in place of a pre-built map.
- Reinforcement learning for decision making.
- Battery-aware mission planning.
- Vision-based package classification.
- ROS 2 integration.
- Digital twin dashboard with live mission monitoring.

11. Learning Outcomes

Upon successful completion of this project, students will be able to:

- Systematically design a mobile robotic platform for a specified mission.
- Evaluate and justify locomotion mechanisms using engineering trade-off analyses.
- Model and simulate complex robotic systems in CoppeliaSim.
- Integrate heterogeneous sensing modalities for environmental perception.
- Develop localization, mapping, planning, and navigation algorithms.

- Design autonomous systems that operate robustly under uncertainty.
- Apply systems engineering principles to the development of a complete autonomous mobile robot.
- Communicate engineering decisions effectively through technical reports and design reviews.